



Why Sodium-ion Batteries Are the Next Big Thing



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A promising next-generation battery technology that may be able to get around some of lithium-ion batteries' drawbacks is sodium-ion batteries. Sodium ions (**Na⁺**) are the charge carriers in sodium-ion batteries (**NIBs**), a kind of rechargeable battery.

They have several benefits over lithium-ion batteries, such as:

- 1. Cost and Abundance:** Compared to lithium, sodium is much more affordable and easier to obtain due to its abundance. Because of this, sodium-ion batteries may become much less expensive, making them more accessible to consumers.
- 2. Thermal Stability:** Compared to lithium-ion batteries, sodium-ion batteries are less likely to experience thermal runaway, a potentially disastrous occurrence. This increases the safety of sodium-ion batteries, especially when used in electric vehicle applications.
- 3. Environmental friendliness:** Compared to lithium mining and processing, sodium mining and processing causes less environmental harm. Hence, sodium-ion batteries are a more environmentally friendly choice.
- 4. Cycle Life:** Sodium-ion batteries can endure a significant number of charge-discharge cycles before experiencing a decline in performance, similar to lithium-ion batteries. This implies that they have a longer useful life.
- 5. High energy density:** As a result of recent developments in sodium-ion battery technology, batteries with energy densities that are on par with or higher than those of lithium-ion batteries

have been created. As a result, sodium-ion batteries are better suited for applications requiring a smaller amount of space because they can store more energy per unit of volume.